

# Monika Santra

Ph.D. Candidate, Computer Science  
The Security of Software (SOS) Group, The Pennsylvania State University  
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## RESEARCH FOCUS

**Software & Systems Security with augmented AI/ML** — binary analysis, vulnerability discovery, control-flow and memory safety, and malware analysis; building scalable, reproducible security tooling.

## EDUCATION

- **The Pennsylvania State University** PA, USA  
*Ph.D. in Computer Science & Engineering; GPA: 4/4* 2022 - present  
*Dissertation: AI-Augmented Static Analysis: Bridging Heuristics and Completeness for Practical Reverse Engineering*  
Advisors: Dr. G. Gary Tan
- **Indian Institute of Technology Roorkee** Uttarakhand, India  
*M. Tech in Computer Science & Engineering; GPA: 9.765/10* 2015 - 2017  
*Dissertation: Design and Implementation of Remote Attestation protocols for servers.*  
Advisors: Dr. Sateesh K. Peddoju (IIT Roorkee) & Dr. Anup K. Bhattacharjee (BARC)
- **Govt. College of Engineering & Textile Technology, Serampore (WBUT)** West Bengal, India  
*B. Tech in Computer Science & Engineering; GPA: 8.86/10* 2010 - 2014

## EXPERIENCE

- The Pennsylvania State University** PA, USA  
*Ph.D. Candidate* 2022 - present
- **Research Experience — Binary Analysis & AI/ML for RE:**
    - DISA — Learning-guided static disassembly & memory structure recovery for CFG**
      - Designed a transformer-based disassembly framework on a shared model architecture that jointly recovers code layout and memory block boundaries on large stripped binaries.
      - Integrated memory block boundaries into a downstream block-level value set analysis-driven CFG recovery tool, improving control-flow via sound and precise indirect call resolution.
    - iResolveX — Hybrid indirect-call resolution via static reasoning and learning-supported refinement**
      - Multi-stage pipeline coupling high-recall target supersets with ML-guided re-ranking and bounded inter-procedural evidence support to produce a new CFG representation, called *p-IndirectCFG*.
      - *p-IndirectCFG*, a practical CFG that ranks indirect edges, allowing controllable pruning (precision vs. recall), consistent thresholds across binaries, and cleaner graphs for IDA/Ghidra/angr.
- Bhabha Atomic Research Centre (BARC)** Maharashtra, India  
*Research Scientist, Grade D* 2019 - 2022
- **Research Experience — Software Security & ML:**
    - **Automated Multilayered Malware Classifier:** Designed, built, and deployed an ensemble malware classifier combining basic/advanced static and dynamic analysis; introduced CFG-driven shallow+deep features; integrated with an Application Whitelisting framework for host-side protection.
    - **Binary Vulnerability Detection via LLVM IR:** Developed a static analysis-driven detector for high-frequency issues (e.g., buffer overflows, memory leaks) in third-party binaries prior to deployment; contributed LLVM IR-level data flow-analysis checks.
    - **SPERTS — Safe Programming Environment for Real-Time Systems:** Developed a Linux-based graphical modeling framework and simulation support for safety-critical I&C systems; implemented complex built-in operators and automated LaTeX/JavaScript report generation.
- Research Scientist, Grade C* 2017 - 2019
- **Research Experience — Systems & Network Security:**
    - **Trusted Systems for Instrumentation & Control (I&C) Network:** Designed and demonstrated the in-house Hardware Reference Monitor (PCIe & USB); delivered **Secure Boot**, **Authentication**, and **Remote Attestation** for host/guest Linux and Windows; built kernel/device drivers & daemons, Application Whitelisting, and a real-time trust dashboard; extended QEMU-KVM for virtualization support; contributed SRAM characterization for PUF-based hardware security.
    - **Advanced Industrial Control Firewall (Modbus/TCP):** Implemented a DPI-based Linux kernel module for application layer-based policy enforcement between field and supervisory networks; shipped a real-time rule monitoring/configuration manager and integrated a high-speed encryption-decryption module to enhance the communication security on the supervisory network side.

- **Research Experience — Network Security:**

- **Design and Implementation of Remote Attestation protocols for servers:** Proposed and implemented a novel remote attestation protocol utilizing the system's underlying hardware and software measurements that successfully addressed the existing low extensibility and openness problems. [Published in IEEE TrustCom 2017]

## PUBLICATIONS

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- **Monika Santra**, Bokai Zhang, Mark Lim, Vishnu Asutosh Dasu, Dongrui Zeng, Gang Tan. “iResolveX: Multi-Layered Indirect Call Resolution via Static Reasoning and Learning-Augmented Refinement.” *USENIX Security*, under review, 2026.
- Peicheng Wang, **Monika Santra**, Mingyu Liu, Cong Sun, Dongrui Zeng, Gang Tan. “DISA: Accurate Learning-based Static Disassembly with Attentions.” *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, 2025 (Cycle A). arXiv:2507.07246
- R. K. Koppanati, **M. Santra**, and S. K. Peddoju. “D<sup>2</sup>4D: Dynamic Deep 4-Dimensional Analysis for Malware Detection.” *IEEE Transactions on Information Forensics and Security (TIFS)*, vol. 20, pp. 2083–2095, 2025. doi: 10.1109/TIFS.2025.3531230
- R. K. Koppanati, **M. Santra**, and S. K. Peddoju. “PotentRegion4MalDetect: Advanced Features from Potential Malicious Regions for Malware Detection.” *Applied Soft Computing Journal*, under review, 2025. arXiv:2507.06723
- ”**M. Santra**, S. K. Peddoju, A. K. Bhattacharjee and A. Khan. Design and Analysis of a Modified Remote Attestation Protocol.”, 2017 IEEE Trustcom/BigDataSE/ICSS, Sydney, NSW, 2017, pp. 578-585

## SKILLS SUMMARY

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- **Computer Languages:** C, C++, Python, Java, Bash

- **Packages:**

GHIDRA, Angr, IDA pro, DWARF, pyelftools, capstone, TREX, Fairseq, Visual Studio Code, Qt Creator, Atom, Eclipse, Pycharm, Arduino Uno, VirtualBox, QEMU, KVM, Radare2, R2pipe, EMBER, SVF, Wireshark, Process Monitor, Kernel debugger, RPM package manager, Windows Driver Framework, Keras, weka, NumPy, Scikit-Learn, Matplotlib, SciPy, Pandas

- **Platforms:** Linux, Windows, Arduino
- **Soft Skills:** Academic Writing, Time Management, Endurance and perseverance towards research, Hardworking
- **Languages:** English, Hindi, and Bengali (Native Language)

## RESEARCH INTERESTS

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Binary analysis, malware analysis, security vulnerability analysis, cyber-resilient trusted & secure system design, system & software security, machine learning in security

## HONORS AND AWARDS

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- Recipient of ACM CCS Doctoral Symposium Travel Grant, 2025
- Recipient of the Dr. Tse-Yun Feng Graduate Student Award (Best RA Award), 2023
- Received ACM-W scholarship, 2017
- Received Travel Grant Scheme from Indian National Academy of Engineering, 2017
- I was the only computer engineer in the country in 2015 who received DAE Graduate Fellowship Scheme (DGFS-2015), Department of Atomic Energy, India, 2015-2017
- Ranked among top 0.1% students all over the country in graduate test in engineering (GATE CS), 2015

## SERVICE

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- Reviewer: *IEEE Transactions on Network Science and Engineering (TNSE)*, 2024
- Organizer: Penn State INSR Industry Day, 2024